

Rajdeep Gill

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EDUCATION

Bachelor of Science in Computer Engineering (Co-op)

University of Manitoba

GPA: 4.45/4.5

Relevant Coursework: Data Structures and Algorithms, Parallel Computing, Computer Architecture, Operating Systems, Embedded Systems, Digital Signal Processing

May 2026

Winnipeg, MB

EXPERIENCE

Robinhood

Software Engineer Intern

- Incoming intern for the Summer 2025 term on the Brokerage team

May 2025 - Sept 2025

Toronto, ON

Biosensors Research Lab - University of Manitoba

Undergraduate Research Student

- Designed and trained a convolutional neural network using Python and PyTorch to predict cell viability from 10,000+ diffraction images, achieving 85% accuracy, demonstrating potential for non-invasive cell viability assessment
- Improved particle diffraction centroid detection success rate by 17% (from 75% to 92%) using computer vision techniques with Python and OpenCV, enhancing data quality for CNN training
- Developed an interactive data visualization tool in Python with Tkinter to process filter, and display model predictions
- Applied unsupervised machine learning (Gaussian Mixture Model) with Python and Scikit-learn to classify cells based on impedance measurements, identifying distinct groups related to cell health

May 2024 - Dec 2024

Winnipeg, MB

Biosensors Research Lab - University of Manitoba

Undergraduate Research Student

- Applied dimensionality reduction using Python and Scikit-learn to analyze and cluster cells based on scattering patterns
- Optimized data analysis pipeline through parallel processing and refactoring existing algorithms, achieving 2.5x speedup
- Developed an automated oscilloscope testing platform with Python and PyVISA, streamlining data collection protocols to enhance measurement speed and ensure repeatability

May 2023 - Aug 2023

Winnipeg, MB

PROJECTS

Google Docs Clone | React, TypeScript, HTML/CSS, Next.js

- Developed a Google Docs clone with real-time collaboration, user authentication, and document sharing
- Utilized Convex to store and sync document data, allowing multiple users to edit in real-time
- Implemented organization-wide document sharing and user permissions to control access to documents

Financial Dashboard | React, TypeScript, HTML/CSS, Next.js, Hono.js, PostgreSQL

- Created an expense and income tracking dashboard with user authentication and real-time data visualization
- Implemented a RESTful API to handle user data and transactions, and a PostgreSQL database to store user information

Machine Learning Classification of Microplastics | Python, PyTorch, OpenCV

- Developed a machine learning model to classify microplastics based on their optical and dielectrophoresis features
- Created a image processing pipeline to extract features, discard low-quality images, and prepare data for training
- Paper accepted at IEEE AP-S/URSI 2025, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features"

Deaf Curling Accessibility Aid | C, ESP-32, Fusion360

- Developed a fully wireless communication system for deaf curlers to receive instructions from their skip during games
- Implemented a custom communication protocol to ensure reliable and low-power communication between devices
- Designed a PCB, 3D-printed enclosure and antenna to ensure durability and portability

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, TypeScript, JavaScript

Frameworks & Libraries: PyTorch, OpenCV, OpenMPI, Tkinter, React, Node, Express, Spring Boot, Flask

Tools: Git, Jenkins, Jira, AWS, Linux, Docker, Firebase, SQL, NoSQL, Excel, REST API

Concepts: Backend Development, Frontend Development, Fullstack Development, Software Engineering, Machine Learning, Networking, CI/CD, Microservices, Infrastructure Engineering, DevOps, Automation, Agile